

Grade 8
Unit 4
Lesson 11 Practice

Name Answer Key
Date _____
Period _____

1. Select all the expressions that are equivalent to the expression $m^{-3}k^5 \cdot mk^{-1}$

- ☒ $\frac{k^5m}{km^3}$
☐ $\frac{k^4}{m}$
☐ $\frac{1}{m^2k^4}$
☐ $m^{-2} \cdot k^2$
☒ $m^{-2}k^4$

2. Select all the expressions that are equivalent to the expression $(3a^{-4}b^{-3})^3 \cdot (-3a)^{-2}$

- ☐ $\frac{27a^{-12}b^{-9}}{9a}$
☒ $\frac{3a^{-13}b^{-9}}{a}$
☐ $\frac{27a^{-14}}{3b^6}$
☐ $\frac{3a^{-7}}{27a^7b^{-2}}$
☐ $(3a^{-4}b^{-1})^3$

For questions 3–10, generate an equivalent expression for each by applying the laws of exponents. Use positive exponents and the fewest factors.

3. $x^5 \cdot (x^3)^{-4}$

$\frac{1}{x^7}$

4. $\frac{(x^7)^3}{x^{-7}}$

x^{28}

5. $(-4x)^{-2} \cdot (-x^3)^2$

$\frac{x^4}{16}$

$$6. \quad \frac{(6xb^{-2})^2}{(2x^{-3}b^{-4})^{-3}}$$

$$\frac{288}{x^7b^{16}}$$

$$7. \quad x^{-7} \cdot x^2 \cdot (x^3)^{-4}$$

$$\frac{1}{x^{17}}$$

$$8. \quad \frac{3x^{-4}y^6}{-18x^3y}$$

$$\frac{y^5}{-6x^7}$$

$$9. \quad \left(\frac{x^4}{3}\right)^{-3} \cdot \frac{x^3}{(4x^5)^{-2}}$$

$$432x$$

$$10. \quad \left(\frac{-2x^0y^4}{5x^{-3}y}\right)^{-2}$$

$$\frac{25}{4x^6y^6}$$